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### Privacy display control apparatus

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#### Abstract

A display apparatus comprises a display device capable of operating in a first mode and a second mode where an image remains visible in both the first mode and the second mode in an on-axis direction and a visibility of the image is reduced in an off-axis direction in the second mode. A control system is capable of selectively operating the display device in the first mode or the second mode and can selectively control luminance of the displayed image in the first mode based on a first function and to selectively control luminance of the displayed image in the second mode based on a second function. High image visibility can be provided for a public mode of operation while in privacy mode of operation a high visual security level may be obtained by means of controlling an image characteristic such as image luminance, contrast and white point.

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## Background/Summary

### TECHNICAL FIELD

[0001] This disclosure generally relates to illumination from light modulation devices, and more specifically relates to control of privacy display.

### BACKGROUND

[0002] Privacy displays provide image visibility to a primary user that is typically in an on-axis position and reduced visibility of image content to a snooper, that is typically in an off-axis position. A privacy function may be provided by micro-louvre optical films that transmit some light from a display in an on-axis direction with low luminance in off-axis positions. However such films have high losses for head-on illumination and the micro-louvres may cause Moiré artefacts due to beating with the pixels of the spatial light modulator. The pitch of the micro-louvre may need selection for panel resolution, increasing inventory and cost.

[0003] Switchable privacy displays may be provided by control of the off-axis optical output.

[0004] Control may be provided by means of luminance reduction, for example by means of switchable backlights for a liquid crystal display (LCD) spatial light modulator. Display backlights in general employ waveguides and edge emitting sources. Certain imaging directional backlights have the additional capability of directing the illumination through a display panel into viewing windows. An imaging system may be formed between multiple sources and the respective window images. One example of an imaging directional backlight is an optical valve that may employ a folded optical system and hence may also be an example of a folded imaging directional backlight. Light may propagate substantially without loss in one direction through the optical valve while counter-propagating light may be extracted by reflection off tilted facets as described in U.S. Pat. No. 9,519,153, which is incorporated by reference herein in its entirety.

[0005] In a known privacy display the privacy mode is provided by the addition of a removable louver film, such as marketed by 3M Corporation, which may not be fitted or removed by users reliably and therefore in practice, is not assiduously attached by the user every time they are outside the office. In another known privacy display the control of privacy mode is electronically activated but control is vested in the user who must execute a keystroke to enter privacy mode.

### BRIEF SUMMARY

[0006] According to the present disclosure, there is provided a display apparatus comprising: a display device arranged to display an image and capable of operating in at least a public mode and a privacy mode, wherein the visibility of the image to an off-axis viewer is reduced in the privacy mode compared to the public mode; and the visibility of the image on-axis is maintained in the privacy mode; and a control system arranged to control the display device, the control system being capable of selectively operating the display device in the public mode or the privacy mode.

[0007] The privacy mode can be used in public places such as cafes or trains in order to enable the primary user to keep working but preventing onlookers or snoopers from being able to see or photograph data from the screen. The public mode can be used when discussing the contents on the

screen with colleagues, for example within the corporate office.

[0008] The display apparatus may be arranged to connect to a network and the control system is arranged to selectively operate the display device in the public mode or the privacy mode in response to the nature of the network.

[0009] Advantageously the operation of the privacy mode may be independent of user preference such that organisations may be provided with control of exposure of private data when used in a public place without the consent of the user.

[0010] The display apparatus may be arranged to receive the identity of the user and the control system is arranged to selectively operate the display device in the public mode or the privacy mode in response to the received identity of the user. Advantageously the privacy operation of the display may be dependent on the identity of the user, so that an organisation may have control of use of public mode of operation of displayed data.

[0011] The display apparatus may comprise a location system arranged to detect the location of the display apparatus and the control system is arranged to selectively operate the display device in the public mode or the privacy mode in response to the detected location. Advantageously, privacy mode may be automatically configured in high risk locations, reducing risk of display of private data and increasing productivity of workers operating in public places.

[0012] The display apparatus may be capable of executing application software and the control system is arranged to selectively operate the display device in the public mode or the privacy mode in response to the context of the application software being executed. The display apparatus may be capable of executing application software for displaying a document, and the control system is arranged to selectively operate the display device in the public mode or the privacy mode in response to the content of the document and/or metadata associated with the document.

Advantageously some high risk applications and data may be provided with increased Visual Security Level in comparison to low risk applications and data.

[0013] The display apparatus may be arranged to receive user input and the control system is arranged to selectively operate the display device in the public mode or the privacy mode in response to the received user input. Advantageously the user may override the privacy operation in allowed circumstances.

[0014] The display apparatus may comprise an ambient light sensor arranged to detect the level of the ambient light. The control system may be arranged to selectively operate the display device in the public mode or the privacy mode in response to the detected level of the ambient light. The control system may be arranged to selectively control any one or more of luminance, contrast, white point and spatial frequency of the image in the privacy mode in response to the detected level of the ambient light. The control system may be arranged to selectively control any one or more of luminance, contrast, white point and spatial frequency of the image in the public mode in response to the detected level of the ambient light. Advantageously the Visual Security Level of operation of the display may be optimised for the illuminated environment. Further image appearance to the primary user during privacy display operation may be increased in response to the type of private image displayed.

[0015] The control system may be arranged to selectively control the display apparatus to display information representing the visibility of the image to an off-axis viewer taking into account the detected level of the ambient light. Advantageously a display user may be provided with information regarding the visual security of the environment to achieve reliable decision making to view confidential data.

[0016] The control system may be arranged to control the display apparatus to display information representing changes to user-controllable parameters capable of reducing the visibility of the image to an off-axis viewer. The control system may be arranged to selectively control any one or more of luminance, contrast, white point and spatial frequency of the image in the privacy mode.

Advantageously the Visual Security Level may be reduced while achieving comfortable viewing of

image data for a primary user that is dependent on the user preference of image viewing.

[0017] The control system may be arranged to selectively control luminance of the displayed image in the public mode and the privacy mode in response to the detected level of the ambient light, using different transfer functions relating levels of luminance to detected levels of the ambient light in the public mode and in the privacy mode. The transfer function in the public mode may relate higher levels of luminance to detected levels of the ambient light than the transfer function in the privacy mode. Advantageously a high visual security level may be provided in privacy mode to snoopers for a wide range of ambient light conditions and a high image visibility may be provided in public mode to users for a wide range of ambient light conditions.

[0018] The control system may be arranged to selectively control luminance of the displayed image in the privacy mode in response to the detected level of the ambient light, in accordance with a transfer function that maintains a relationship  $Y_0/I \geq 1$  lux/nit, where  $Y_0$  is the maximum output luminance of the display apparatus measured in nits and  $I$  is the detected ambient light level measured in lux. Advantageously high visual security level is observed for varying ambient light levels and desirable image visibility is provided to the display user.

[0019] The control system may be arranged to selectively control luminance of the displayed image in the privacy mode in response to the detected level of the ambient light, in accordance with a transfer function that maintains a relationship  $Y_0 \leq Y_{0.\text{sub.lim}}$ , where  $Y_{0.\text{sub.lim}}$  is given by the equation:

$$[00001] Y_{0.\text{lim}} = \frac{R_{\theta} + V_{\text{lim}} * K}{(V_{\text{lim}} - 1) * P_{\theta}}$$

where  $V_{\text{sub.lim}}$  has a value of 10,  $R_{\theta}$  is the reflected ambient illuminance at an observation angle  $\theta$  of 45 degrees laterally and zero degrees in elevation from the normal to the display apparatus,  $K_{\theta}$  is the display black state luminance at the observation angle, and  $P_{\theta}$  is the relative luminance at the observation angle  $\theta$  compared to the maximum display output luminance  $Y_0$ .

[0020] The control system may be arranged to control the display apparatus to display information representing the visibility of the image to an off-axis viewer. Advantageously the Visual Security Level of the display apparatus may be optimised.

[0021] The display apparatus may be arranged to receive time data input and ambient light sensor input and the control system is arranged to selectively operate the display device in the public mode or the privacy mode in response to the received time input and ambient light sensor input in order to reduce light pollution generated by the display apparatus.

[0022] Advantageously at night time in a private location such as a bedroom, the light level and angular light output range may be reduced so that the display is less distracting to others in the environment.

[0023] The display apparatus may be arranged where the content and/or metadata includes the security status of the transfer protocol of the document and determining it to be Hypertext Transfer Protocol or Hypertext Transfer Protocol Secure.

[0024] Advantageously the display apparatus, when accessing a web page or similar document may use meta data associated with that document or associated with the transfer process, for example if it is set to be transferred using a secure or an open transfer protocol to determine if the display apparatus should be set to privacy mode and therefore maintain good data privacy.

[0025] In this document, public and privacy refer to the mode of the display rather than the nature of the location. For example privacy (display) mode is typically selected in public places such as coffee shops and public (display) mode is typically selected in private location such as at home.

[0026] Any of the aspects of the present disclosure may be applied in any combination.

[0027] Embodiments of the present disclosure may be used in a variety of optical systems. The embodiments may include or work with a variety of projectors, projection systems, optical components, displays, microdisplays, computer systems, processors, self-contained projector systems, visual and/or audio-visual systems and electrical and/or optical devices. Aspects of the present disclosure may be used with practically any apparatus related to optical and electrical

devices, optical systems, presentation systems or any apparatus that may contain any type of optical system. Accordingly, embodiments of the present disclosure may be employed in optical systems, devices used in visual and/or optical presentations, visual peripherals and so on and in a number of computing environments.

[0028] Before proceeding to the disclosed embodiments in detail, it should be understood that the disclosure is not limited in its application or creation to the details of the particular arrangements shown, because the disclosure is capable of other embodiments. Moreover, aspects of the disclosure may be set forth in different combinations and arrangements to define embodiments unique in their own right. Also, the terminology used herein is for the purpose of description and not of limitation.

[0029] These and other advantages and features of the present disclosure will become apparent to those of ordinary skill in the art upon reading this disclosure in its entirety.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

[0030] Embodiments are illustrated by way of example in the accompanying FIGURES, in which like reference numbers indicate similar parts, and in which:

[0031] FIG. 1 is a schematic diagram illustrating a front view of a privacy display comprising a privacy control system operating in privacy mode with a first visual security level;

[0032] FIG. 2 is a schematic diagram illustrating a look-down perspective of a privacy display comprising a privacy control system operating in privacy mode with a first visual security level;

[0033] FIG. 3 is a schematic diagram illustrating a front view of a privacy display comprising a privacy control system operating in public mode;

[0034] FIG. 4 is a schematic diagram illustrating a look-down perspective of a privacy display comprising a privacy control system in public mode;

[0035] FIG. 5A illustrates a flowchart for a privacy display comprising a privacy control system;

[0036] FIG. 5B illustrates a flowchart of the privacy control system of FIGS. 1-2;

[0037] FIG. 6A is a schematic graph illustrating variation of output luminance with viewing angle for a typical collimated backlight arranged to cooperate with a switchable retarder to provide high visual security level to a wide range of snooper locations;

[0038] FIG. 6B is a schematic graph illustrating variation of Visual Security Level with off-axis relative luminance of a switchable privacy display operating in privacy mode;

[0039] FIG. 6C is a schematic graph illustrating variation of reflectivity with polar viewing angle for two types of privacy display;

[0040] FIG. 6D is a schematic graph illustrating variation of Visual Security Level with polar viewing angle for the two types of privacy display of FIG. 6C;

[0041] FIG. 7A is a schematic graph illustrating transfer functions between head-on display luminance and ambient illuminance;

[0042] FIG. 7B is a schematic graph illustrating transfer functions between the ratio of measured ambient illuminance to head-on display luminance and ambient illuminance;

[0043] FIG. 7C is a schematic graph illustrating the variation of perceived privacy with visual security level;

[0044] FIG. 8A is a schematic diagram illustrating a front view of a privacy display comprising a privacy control system operating in privacy mode with a second visual security level;

[0045] FIG. 8B is a schematic diagram illustrating a look-down perspective of a privacy display comprising a privacy control system operating in privacy mode with a second visual security level;

[0046] FIG. 9 is a schematic graph of CIE 1931 x-y coordinates illustrating variation in image white point with Visual Security Level;

[0047] FIG. 10 illustrates a flowchart of the privacy control system of FIGS. 1-2 and FIGS. 3-4;  
[0048] FIG. 11A illustrates a flowchart for control of a privacy display by means of calculation of Visual Security Level;  
[0049] FIG. 11B illustrates a flowchart for control of a privacy display by means of calculation of Visual Security Level and user feedback on preferred improvements;  
[0050] FIG. 11C illustrates a flowchart for control of a privacy display comprising an allowable visual security level for a working document;  
[0051] FIG. 11D illustrates a flowchart for control of a privacy display comprising an allowable visual security level for a password protected document;  
[0052] FIG. 12A and FIG. 12B are schematic diagrams illustrating a front view of a privacy display comprising a privacy control system for adjustment of image visibility in public mode;  
[0053] FIG. 13 is a schematic graph illustrating variation of image visibility with off-axis relative luminance of a switchable privacy display operating in public mode;  
[0054] FIG. 14 illustrates a flowchart for control of a switchable privacy display operating in public mode by means of calculation of image visibility for off-axis users;  
[0055] FIG. 15 is a schematic diagram illustrating in front perspective view a switchable directional display apparatus comprising a switchable directional backlight and a switchable liquid crystal retarder;  
[0056] FIG. 16A is a schematic diagram illustrating in perspective side view an arrangement of a switchable liquid crystal retarder comprising a passive negative C-plate compensation retarder in a privacy mode of operation;  
[0057] FIG. 16B is a schematic diagram illustrating in side view propagation of output light from a spatial light modulator through the optical stack of FIG. 15 in a privacy mode of operation;  
[0058] FIG. 16C is a schematic graph illustrating the variation of output luminance with polar direction for the transmitted light rays in FIG. 16B;  
[0059] FIG. 16D is a schematic diagram illustrating in top view propagation of ambient illumination light through the optical stack of FIG. 15 in a privacy mode of operation;  
[0060] FIG. 16E is a schematic graph illustrating the variation of reflectivity with polar direction for the reflected light rays in FIG. 16D;  
[0061] FIG. 17A is a schematic diagram illustrating in perspective side view an arrangement of a switchable retarder in a public mode of operation wherein the switchable retarder comprises a switchable liquid crystal layer with homeotropic alignment and a passive C-plate compensation retarder;  
[0062] FIG. 17B is a schematic diagram illustrating in side view propagation of output light from a spatial light modulator through the optical stack of FIG. 15 in a public mode of operation;  
[0063] FIG. 17C is a schematic graph illustrating the variation of output luminance with polar direction for the transmitted light rays in FIG. 17B;  
[0064] FIG. 17D is a schematic diagram illustrating in top view propagation of ambient illumination light through the optical stack of FIG. 15 in a public mode of operation; and  
[0065] FIG. 17E is a schematic graph illustrating the variation of reflectivity with polar direction for the reflected light rays in FIG. 17D.

#### DETAILED DESCRIPTION

[0066] Terms related to privacy display appearance will now be described.

[0067] A private mode of operation of a display is one in which an observer sees a low contrast sensitivity such that an image is not clearly visible. Contrast sensitivity is a measure of the ability to discern between luminances of different levels in a static image. Inverse contrast sensitivity may be used as a measure of visual security, in that a high visual security level (VSL) corresponds to low image visibility.

[0068] For a privacy display providing an image to an observer, visual security may be given as:

[00002]  $V = (Y + R) / (Y - K)$  eqn. 1 [0069] where V is the visual security level (VSL), Y is the luminance of the white state of the display at a snoopers viewing angle, K is the luminance of the black state of the display at the snoopers viewing angle and R is the luminance of reflected light from the display.

[0070] Panel contrast ratio is given as:

[00003]  $C = Y / K$  eqn. 2 [0071] so the visual security level may be further given as:

[00004]  $V = (P \cdot L + I \cdot \rho) / (P \cdot (L - L / C))$  eqn. 3 [0072] where off-axis relative luminance, P is typically defined as the percentage of head-on luminance, L at the snoopers angle and the display may have image contrast ratio C and the surface reflectivity is  $\rho$ .

[0073] The off-axis relative luminance, P is sometimes referred to as the privacy level. However, such privacy level P describes relative luminance of a display at a given polar angle compared to head-on luminance, and is not a measure of privacy appearance.

[0074] The display may be illuminated by Lambertian ambient illuminance I. Thus in a perfectly dark environment, a high contrast display has VSL of approximately 1.0. As ambient illuminance increases, the perceived image contrast degrades, VSL increases and a private image is perceived.

[0075] For typical liquid crystal displays the panel contrast C is above 100:1 for almost all viewing angles. allowing the visual security level to be approximated to:

[00005]  $V = 1 + I \cdot \rho / (C \cdot P \cdot L)$  eqn. 4

[0076] In the present embodiments, in addition to the exemplary definition of eqn. 4, other measurements of Visual Security Level, V may be provided, for example to include the effect on image visibility to a snoopers of snoopers location, image contrast, image colour and white point and subtended image feature size. Thus the Visual Security Level may be a measure of the degree of privacy of the display but may not be restricted to the parameter V.

[0077] In comparison to privacy displays, desirably wide angle displays are easily observed in standard ambient illuminance conditions. One measure of image visibility is given by the contrast sensitivity such as the Michelson contrast which is given by:

[00006]  $M = (I_{\max} - I_{\min}) / (I_{\max} + I_{\min})$  eqn. 5 [0078] and so:

[00007]  $M = ((Y + R) - (K + R)) / ((Y + R) + (K + R)) = (Y - K) / (Y + K + 2 \cdot R)$  eqn. 6

[0079] Thus the visual security level (VSL), V is equivalent (but not identical to)  $1/M$ . In the present discussion, for a given off-axis relative luminance, P the wide angle image visibility, W is approximated as

[00008]  $W = 1 / V = 1 / (1 + I \cdot \rho / (C \cdot P \cdot L))$  eqn. 7

[0080] It would be desirable to provide control of a switchable privacy display that is determined by user environment and operating function.

[0081] FIG. 1 is a schematic diagram illustrating a front view of a privacy display apparatus 100 comprising a privacy control system 500 operating in privacy mode with a first visual security level.

[0082] It may be desirable for the display apparatus 100 to enter privacy mode automatically so that managers may reliably enforce the implementation of a corporate policy 240. The control system 500 is capable of selectively operating the display apparatus 100 in the public mode or the privacy mode.

[0083] Display system 200 may comprise privacy mode capable display apparatus 100 and control system 500. The display apparatus comprises a display apparatus 100 arranged to display an image and capable of operating in at least a public mode and a privacy mode, wherein the visibility of the image to an off-axis viewer is reduced in the privacy mode compared to the public mode and the visibility of the image to the primary user in an on-axis position remains visible in both the privacy and public modes. Examples of suitable types of display device are described further below.

[0084] Means to determine privacy mode operation will now be described.

[0085] For a head-on user in typical ambient illuminance environments, desirably the display apparatus **100** provides a displayed image **101** that has a luminance to achieve high image visibility, W in both privacy and public modes of operation.

[0086] The display system **200** may also comprise inputs related to desirable circumstances to provide privacy images, or conversely by undesirable circumstances to provide public images. Such desirable and undesirable circumstances may be determined by policy **240** that is provided for example by a corporate policy, government policy, medical ethical policy or by user preference settings.

[0087] Time signal **262** may be provided to correlate with expected working or usage patterns. For example, at night time, privacy mode may be automatically enabled to achieve reduced stray light to other users. Further privacy operation can be coordinated with known calendar entries for the user, for example to coincide with known travel arrangements.

[0088] Network ID **201** may be provided to wireless network **202** by means of communication **204**. The network ID **201** and wireless network **202** may be determined to be an external network and privacy mode enabled. The display apparatus **100** is arranged to connect to a network and the network name or identity ID **201** can be used by the control system **500** to recognise the location of the display apparatus **100**. For example if a known network with the attribute of “home” or “office” is detected, the display apparatus may be controlled to enable the public display mode. Conversely unknown network connections or no network connection may be recognised as inferring that the privacy display mode should be activated automatically.

[0089] If a private network such as a cell phone connection is in use and the user is not in a private location privacy mode may be enabled for example. Further, if the display is operated in “airplane mode” privacy operation may be automatically enabled.

[0090] Location **206** may be determined by measurement **260** from GPS **258**. Some locations may be defined as requiring privacy mode operation. The location may further be detected from data on the geographical location of the cell phone base station tower provided by the network ID **201** that the mobile data connection to the device is using. The variation in the position of the GPS sensor **206** signal with time may be used by the control system **500** to determine that the device is moving beyond a certain speed and therefore to infer that the display apparatus **100** is for example on a train and that the display apparatus **100** should be controlled to enter privacy mode. Similarly, GPS sensor **206** may be used to determine the altitude of the device and thereby to infer that the device is on an airplane and that the display apparatus **100** should automatically switch to privacy mode. When GPS sensor **206** is unavailable and when the device is connected to a corporate network with Network ID **201** via a mobile data connection e.g. 3G, 4G or 5G, the doppler effect on the radio signals may be used to determine the speed of movement and thereby to infer that the device is in motion such as on a train and therefore switch automatically to privacy mode.

[0091] The control system **500** may be arranged to selectively operate the display apparatus **100** in the public mode or the privacy mode in response to the detected level of the ambient light. The ambient light level may be detected by an Ambient Light Sensor **232** such as a photodiode which may have a photopic filter or a photopic light response current or voltage or digital value.

[0092] Some types of display have multiple optical effects to improve privacy performance, with exemplary optical effects described below. If more than one privacy optical effect is available, the mode that gives the widest viewing freedom for the primary user while still maintaining adequate Visual Security Level at the ambient light level experienced can be selected by the control system **500**. Advantageously privacy is protected, and user productivity is maintained.

[0093] Visible applications **274** may be analysed to determine whether content of images **101** is suitable for public display. If private application **276** is visible on the display, then privacy mode operation is provided.

[0094] Visible applications **276** may also be provided with text **224** and images that have feature sizes (such as font sizes) that have controlled modulation transfer functions when the applications



are operated in privacy mode. For example in privacy mode of operation text size may be reduced. [0095] Camera **230** and user identification system (not shown) may be used to determine that the user is not permitted to operate the display in public mode. Thus the control system may selectively operate the display device in the public mode or the privacy mode in response to the received identity of the user.

[0096] The display apparatus **100** may be arranged to receive the identity of the user, for example by means of face detection from images provided by the camera **230**. Other known means of identifying the user including but not limited to fingerprint or retina scan may also be used. External command **243** may be provided by a remote user to lock the display to privacy mode, for example to control privacy to defined groups of display users.

[0097] Airplane mode **270** may be selected, indicating that the device is in a public space and privacy operation may be required by the policy **240**. Further, airplane viewing conditions at night time may be a specifically demanding snooping environment, and so the control system **500** may trigger additional privacy enhancement including contrast reduction, white point shift and spatial frequency changes as will be described further below.

[0098] Local command to over-ride public display operation may be provided by user override **46** by means of key **47** press for example. It may be desirable for the user to override the default privacy settings under conditions allowed by policy **240**. The control system **500** selectively operates the display apparatus **100** in the public mode or the privacy mode in response to the received user override **46**. The user may determine that whilst in flight or in a train his specific location is not vulnerable to being overlooked and so it may be preferable to operate the display apparatus **100** in public mode. Advantageously in public mode the display apparatus **100** may have greater image uniformity and viewing freedom for the primary user as well as being visible from multiple viewing locations.

[0099] Visual security level indicator **280** may be provided on the display which is a measure of the privacy level achieved. In the illustrative example of FIG. **1**, the indicator **282** may be an amber privacy warning that indicates there may be some residual image visibility to an off-axis snooper. When switched in to privacy mode the control system **500** may be arranged to control the display apparatus **100** to display image **101** with information such as indicator **280** representing the visibility of the image to an off-axis viewer, for example to provide Visual Security Level, V. Advantageously the user or their supervisor may be confident in the privacy level being achieved in the specific environment in which they are operating.

[0100] The display appearance in privacy mode as seen by a snooper will now be described together with further inputs for the control of visual security level.

[0101] FIG. **2** is a schematic diagram illustrating a look-down perspective of a privacy display comprising a privacy control system **500** operating in privacy mode with a first visual security level.

[0102] As will be described below, off-axis privacy may be provided by control of off-axis luminance, reflectivity and image contrast of the image **103** provided by a switchable privacy display apparatus **100** to an undesirable snooper.

[0103] In operation in privacy mode, a limited output cone angle **402C** that is typically centred on the optical axis **199** that is a typically a surface normal to the display apparatus **100** is provided. Off-axis luminance is reduced. Ambient light sources **604** illuminate the display surface with light rays **610**. Reflected light **606** from the display provides increased visual security level, V as described above.

[0104] Some light rays **608** may be incident on Ambient Light Sensor, ALS **232**. The ALS **232** may be a separate element or may be incorporated in the camera **230** detection system.

[0105] Ambient illuminance detection **234** provides a calculation of ambient illuminance and is input into the control system **500**. VSL calculation **250** is used to determine desirable display setting characteristics and output to display control **710**. The display control **710** may control

display luminance setting **278** and may be further used to provide visual security level indicator **280** level **282**. Display control **710** is further described below in relation to an example of a privacy display.

[0106] Features of the embodiment of FIG. **2** not discussed in further detail may be assumed to correspond to the features with equivalent reference numerals as discussed above, including any potential variations in the features.

[0107] Advantageously a corporate privacy policy may be enforced when the display apparatus **100** is displaying data and is not in the office or at home. In this display apparatus the security of data on the display apparatus **100** is not reliant on the user fitting a louver screen or on the user pressing a privacy button or key. Advantageously corporate privacy policy may be reliably enforced without dependence on the diligence of the user.

[0108] It would be desirable to provide high image visibility to multiple display users in protected environments such as office environments or the home.

[0109] FIG. **3** is a schematic diagram illustrating a front view of a privacy display comprising a privacy control system operating in public mode; and FIG. **4** is a schematic diagram illustrating a look-down perspective of a privacy display comprising a privacy control system in public mode.

[0110] In comparison to FIG. **1**, conditions that individually or in combination provide allowable public display use will now be described. In public display mode, the output light cone **402D** from the switchable privacy display apparatus **100** may be adjusted to be larger than in privacy mode, such that off-axis display luminance is increased.

[0111] Time signal **262** may be provided to correlate with expected working or usage patterns, for example, in usual working hours.

[0112] The wireless ID **201** and wireless network **202** may be determined to be an internal network.

[0113] Certain office locations or other allowed spatial locations for the display system **200** may be permitted.

[0114] Visible applications **274** may be analysed to determine whether content is suitable for public display.

[0115] All applications **274** may also be provided with desirable text **224** size and image properties with high image contrast.

[0116] Camera **230** and user identification system (not shown) may be used to determine that all visible users are permitted.

[0117] Visual security level indicator **280** may be provided on the display which is a measure of the privacy level achieved. In the illustrative example of FIG. **3**, the indicator **282** may be a red privacy warning that indicates there is high image visibility to an off-axis snooper.

[0118] Control of a switchable privacy display will now be further described.

[0119] Features of the embodiment of FIG. **3** and FIG. **4** not discussed in further detail may be assumed to correspond to the features with equivalent reference numerals as discussed above, including any potential variations in the features.

[0120] FIG. **5A** illustrates a flowchart for a privacy display comprising a privacy control system. Thus a control system **500** is arranged to control the display apparatus **100**.

[0121] FIG. **5B** illustrates a flowchart of the privacy control system **500** that selectively operates the display apparatus **100** in the public mode or the privacy mode in response to the nature of the network to which it is connected or the output of sensors or corporate privacy policy **240**. Thus FIG. **5B** illustrates further the control system **500** that selectively operates the display apparatus **100** in the public mode or the privacy mode in response to the detected location.

[0122] Features of the embodiment of FIG. **5A** and FIG. **5B** not discussed in further detail may be assumed to correspond to the features with equivalent reference numerals as discussed above, including any potential variations in the features.

[0123] Switching between exemplary privacy and public luminance profiles will now be described.

[0124] FIG. **6A** is a schematic graph illustrating variation of output luminance with viewing angle

for a typical collimated backlight arranged to cooperate with plural retarders **300** to provide high visual security level from a wide range of snoopers locations. Features of the embodiment of FIG. **6A** not discussed in further detail may be assumed to correspond to the features with equivalent reference numerals as discussed above, including any potential variations in the features.

[0125] FIG. **6A** illustrates a desirable luminance profile **486** of a backlight **20** operated in privacy mode for use with the switchable liquid crystal retarder **300** of FIG. **15** in privacy mode.

[0126] The profile **486** is modified by switchable liquid crystal retarder **300** to provide an illustrative profile **490** that advantageously achieves an off-axis relative luminance of less than 0.5% at 45 degrees lateral angle and zero degrees elevation, that may be the target snoopers viewing polar locations **27L**, **27R** illustrated in FIG. **16C** for example.

[0127] Control of Visual Security Level will now be further described.

[0128] FIG. **6B** is a schematic graph illustrating variation of Visual Security Level with off-axis relative luminance of a switchable privacy display operating in privacy mode and with reference to the privacy display of FIG. **15** and with reference to FIGS. **16A-16E** and FIGS. **17A-17E**, as exemplary embodiments of a switchable privacy display **100**. Features of the embodiment of FIG. **6B** not discussed in further detail may be assumed to correspond to the features with equivalent reference numerals as discussed above, including any potential variations in the features.

[0129] FIG. **6B** illustrates the profiles for visual security level,  $V$  (calculated in each illustrative embodiment from eqn. 4, above) and with the illustrative embodiments as illustrated in TABLE 1 for varying privacy levels achieved at the target snoopers viewing locations **26L**, **26R**. Display reflectivity of 30% or more may be achieved for displays comprising reflective polariser **302**, while display reflectivity of approximately 5% may be achieved for displays not comprising reflective polariser **302**.

TABLE-US-00001 TABLE 1 Display Ambient Head-on Visual Security reflectivity, illuminance, luminance, Level,  $V$  @ 45° Profile  $\rho$  (%) I (lux) L (nits) lateral angle, 458 — 30 500 200 48.7 — 30 500 300 32.8 450 30 300 300 20.1 452 30 150 300 10.5 — 5 500 100 16.9 — 5 500 200 9.0 454 5 300 300 4.2 456 5 150 300 2.6

[0130] At 0.5% privacy level, various Visual Security Level points **458** may be provided depending on display structure, ambient illuminance and display luminance. The present embodiments further provide indicator **280** for display of Visual Security Level that may be provided for example by means of traffic light indicators.

[0131] The variation of display reflectivity with viewing angle will now be described.

[0132] FIG. **6C** is a schematic graph illustrating variation of reflectivity with polar viewing angle for two types of privacy display. Profile **820** illustrates the variation of reflectivity for an illustrative embodiment of FIGS. **16D-E** and profile **822** illustrates the variation of reflectivity for an embodiment without the reflective polariser **302** of FIG. **15** and FIG. **16A**. Both profiles include Fresnel reflectivity at the outer polariser **318** and thus increase at high polar angles.

[0133] The variation of visual security level,  $V$  with viewing angle will now be described.

[0134] FIG. **6D** is a schematic graph illustrating variation of visual Security Level with polar viewing angle for the two types of privacy display of FIG. **6C**. VSL profile **824** illustrates an output for a display of the type of FIG. **15** with reflective polariser **302**, and VSL profile **826** illustrates an output for the display of FIG. **15** with the reflective polariser **302** omitted. VSL profiles are illustrated for the same ambient illuminance,  $I$ . The limit  $V_{sub.lim}$  above which no image visibility is present is described further below. The angular range **825** of snoopers locations for the profile **824** is thus greater than the angular range **827** for the profile **826**. The reflective polariser **302** achieves above threshold visual security level over a wider polar range, advantageously achieving increased protection from snoopers. Further, head-on luminance may be increased for a given ambient illuminance, increasing image visibility for the display user.

[0135] Selective control of the relationship between desirable display luminance and ambient light illuminance will now be described.

[0136] FIG. 7A is a schematic diagram illustrating exemplary transfer functions **802**, **804** between display head-on luminance and detected ambient illuminance; and FIG. 7B is a schematic graph illustrating transfer functions between the ratio of measured ambient illuminance to head-on display luminance and ambient illuminance.

[0137] Variations **802**, **803** are illustrative profiles for public mode of operation and variations **804**, **805** are illustrative profiles for privacy mode of operation.

[0138] The control system **500** is arranged to selectively control luminance of the displayed image in the public mode and the privacy mode in response to the detected level of the ambient light, in accordance with different transfer functions **802**, **804** relating levels of luminance to detected levels of the ambient light in the public mode and in the privacy mode respectively. The transfer function **802** in the public mode relates higher levels of luminance to detected levels of the ambient light than the transfer function **804** in the privacy mode.

[0139] When operating in the privacy mode the privacy transfer function **804** is selected and the control system uses the measured ambient light level to control the display luminance so that a desirable Visual Security Level,  $V$  at at-least one off-axis snoopers observation angle is provided for different ambient illumination levels. Advantageously display security may be maintained in different or varying lighting conditions.

[0140] When operating in the public mode the public transfer function **802** may be selected to provide a desirable image visibility,  $W$  for different ambient illumination levels. Advantageously display visibility may be maintained in different lighting conditions for off-axis observers.

[0141] Desirable limits for head-on luminance of the display operating in privacy mode will now be described.

[0142] FIG. 7C is a schematic graph illustrating the variation **806** of perceived visual security with visual security level,  $V$  at an observation angle  $\theta$ . Visual security level  $V$  is a measured quantity of any given display and varies with polar viewing angle.

[0143] By comparison with visual security level  $V$ , perceived visual security is a subjective judgement of the visibility of a displayed privacy image arising from the human visual system response at the observation angle.

[0144] In operation it has been discovered that above a threshold limit  $V_{sub.lim}$  of visual security level  $V$  then no image information is perceived. This transition in the perceived visibility with changes in the visual security level  $V$  is very rapid, as shown by the steepness of the graph in FIG. **8** around the threshold limit  $V_{sub.lim}$ . That is, as the visual security level  $V$  increases, initially the perceived visibility degrades only gradually and the image is essentially viewable. However, on reaching the threshold limit  $V_{sub.lim}$ , the perceived image rapidly ceases to be visible in a manner that is in practice surprising to watch.

[0145] In observation of the surprising result, for a text document image that is of concern for privacy applications it was found that the perceived image seen by a snooper rapidly ceased to be visible for  $V$  of 10. In the region **810** for values of  $V$  above 10, all the displayed text had zero visibility. In other words the perceived text rapidly ceases to be visible in a manner that is in practice surprising to watch for  $V$  of 10 or greater.

[0146] In regions **812** below  $V_{sub.lim}$  text was visible with low contrast and in region **814** below  $V'$  text was clearly visible.

[0147] It would be desirable to maximise head-on display luminance to achieve high image visibility to the primary display user. It would be further desirable to achieve high image security level for a snooper at the observation angle. The selective control of the head-on luminance will now be described in further detail.

[0148] For an observation angle  $\theta$ , the maximum display output luminance  $Y_0$  (typically the head-on luminance) is prevented from exceeding a luminance limit  $Y_{0.sub.lim}$  at which the visual security level  $V$  is above the threshold limit  $V_{sub.lim}$  so that the image is not perceived as visible at that observation angle  $\theta$ , the luminance limit  $Y_{0.sub.lim}$  being given by:

$$[00009] \ Y0_{\text{lim}} = \frac{R + V_{\text{lim}} * K}{(V_{\text{lim}} - 1) * P} \quad \text{eqn. 8}$$

where  $R\theta$  is the reflected ambient illuminance at the observation angle  $\theta$ ,  $K\theta$  is the display black state luminance at the observation angle, and  $P\theta$  is the relative luminance at the observation angle  $\theta$  compared to the maximum display output luminance  $Y0$  (typically the head-on luminance and is measured in nits). For display reflectivity  $p\theta$  and a Lambertian illuminant with illuminance  $I\theta$  measured in lux that is reflected by the display at the observation angle, the luminance limit  $Y0_{\text{sub.lim}}$  is also given by:

$$[00010] \ Y0_{\text{lim}} = \frac{*I + V_{\text{lim}} * K}{(V_{\text{lim}} - 1) * P} \quad \text{eqn. 9}$$

[0149] Since the illuminance  $I\theta$  is dependent on the amount of ambient light, the luminance of the display apparatus may be controlled by the control system **500** in accordance with these relationships. Specifically, the privacy transfer function **804** used by the control system **500** as described above may be selected to maintain the relationship  $Y0 \leq Y0_{\text{sub.lim}}$  in order that the image is not perceived as visible at a desired observation angle  $\theta$ , for example at an observation angle  $\theta$  of 45 degrees laterally and zero degrees in elevation from the normal to the display apparatus.

[0150] Subject to that limit, the luminance is preferably as high as possible in order to optimise the performance for the head-on view. Accordingly, the privacy transfer function **804** used by the control system **500** as described above may additionally be selected to maintain the relationship  $Y0/I\theta \geq 1$  lux/nit as illustrated by the profile **805** in FIG. **8**. The illuminance  $I\theta$  may be the sensed ambient illuminance that is averaged from the illuminated scene.

[0151] Advantageously a display may be provided that has high image security to off-axis snoopers while achieving high image visibility for the head-on user for different illuminance levels may be provided.

[0152] FIG. **8A** is a schematic diagram illustrating a front view of a privacy display comprising a privacy control system operating in privacy mode with a second visual security level; and FIG. **8B** is a schematic diagram illustrating a look-down perspective of a privacy display comprising a privacy control system operating in privacy mode with a second visual security level.

[0153] In operation, ambient light sensor **232** may provide a measurement of ambient illuminance. In comparison to FIGS. **1-2**, the illuminance of light rays **608** may be lower, and thus the ratio  $I/L$  reduces, reducing Visual Security Level for a given head-on luminance. The control system **500** may be arranged to perform calculation of Visual Security Level, and reduce display luminance,  $L$  to increase VSL to desirable levels by means of display control **710**. Thus the luminance **278** seen by the head-on user reduces. Such an improvement may be highlighted by a change in the indicator **280** from red to green or amber, depending on desirable levels of Visual Security Level.

[0154] The desirable levels of Visual Security Level for display operation may be set by policy **240**, and said policy may vary said desirable levels of Visual Security Level related to the nature of the viewing environment. For example, in high risk environments, the allowable head-on luminance **278** may be lower than in low risk environments. At threshold environments, for example in very low levels of ambient illumination **604** illuminance,  $I$  then viewing of certain content may be prohibited.

[0155] Features of the embodiment of FIG. **8A** and FIG. **8B** not discussed in further detail may be assumed to correspond to the features with equivalent reference numerals as discussed above, including any potential variations in the features.

[0156] FIG. **9** is a schematic graph of CIE 1931 x-y coordinates illustrating variation in image white point with Visual Security Level.

[0157] In operation, the white point of the display may be adjusted along an illustrative locus **602** in dependence on the calculated Visual Security Level.

[0158] Thus the display when operating in privacy mode may be arranged with white point **605**

when the calculated VSL is high, and the white point may be adjusted to locations **606**, **608** in dependence on the calculated VSL. Thus the display may appear more yellow as the VSL reduces. [0159] Advantageously the image contrast of black on yellow images is lower than for white on black images, thus reducing image visibility to the off-axis snooper.

[0160] A colour adjusted VSL parameter may further be defined based on the white point selected, so that the indicator **280** may be adjusted accordingly.

[0161] Colour adjustment may also be automatically made when a device is adjusted to airplane mode and is operating with privacy images for example.

[0162] Features of the embodiment of FIG. **9** not discussed in further detail may be assumed to correspond to the features with equivalent reference numerals as discussed above, including any potential variations in the features.

[0163] Further description of the control of a privacy display will now be described.

[0164] FIG. **10** illustrates a flowchart of the privacy control system of FIGS. **1-2** and FIGS. **3-4**.

[0165] The display operating environment **261** may include (but is not limited to) network ID **201**, Date/Time **262**, GPS **206** data, Ambient Light Sensor **232** detection and Airplane mode **270** setting.

[0166] Corporate privacy policy **240** may include definitions under which the display should be operated in privacy mode including time and location; documents and applications; and Visual Security Level specifications.

[0167] Other inputs may include display design parameters **272** and information on viewed documents and applications **274**.

[0168] Data processor **290** is used to analyse display operating environment **261**, display design parameters **272**, viewed documents and applications **274** and compare against corporate privacy policy **240**. The output determines whether to operate the display in privacy or public mode such that switch **292** is set for privacy or public mode operation based on data processor **290** output.

[0169] In the case of privacy mode operation the settings to apply to the display apparatus **100** using display control system **710** and images **101** using image control system **296** in order to achieve desirable Visual Security Level are provided. Further indication of Visual Security Level using indicator **280** may be provided.

[0170] In the case of public mode operation, the appropriate illumination control including cone angle change by display control system **710** and luminance using LED driver **715** are provided to the display apparatus **100**.

[0171] The controller **500** may continue to monitor the status of the display operating environment **261** and appropriate changes in policy **240** and adjust display apparatus **100** and images **101** appropriately to maintain the target Visual Security Level.

[0172] Advantageously the control system **500** may enable the Visual Security Level, that may be the Visual Security Level to be reliably calculated and compared to a corporate policy **240** level set for the device's current environment. The Visual Security Level may be adjusted to the level required for the display apparatus **100** environment so that the primary user retains optimal viewing freedom and comfort consistent with achieving the prescribed corporate privacy policy privacy level.

[0173] Features of the embodiment of FIG. **10** not discussed in further detail may be assumed to correspond to the features with equivalent reference numerals as discussed above, including any potential variations in the features.

[0174] The control of a privacy display in response to a measured Visual Security Level will now be described.

[0175] FIG. **11A** illustrates a flowchart for control of a privacy display by means of calculation of Visual Security Level.

[0176] The display apparatus **100** may be provided wherein the control system **500** is arranged to selectively control any one or more of luminance of the display **100** or contrast, white point and spatial frequency of the image **101** in the privacy mode in response to the measured level of the

ambient light using Ambient Light Sensor **232**.

[0177] A luminance evaluation system **904** may be provided to determine output luminance to the head-on user and snooper based on measurement of LED current and display apparatus **100** calibrated characteristics for example, that may be provided as display design parameters **272**.

[0178] An image analysis engine **906** may provide information on images **101** including but not limited to text sizes, image spatial resolution, colour, and contrast.

[0179] Such data may be collected by VSL input data unit **910** and Visual Security Level calculated accordingly in VSL calculator **914**.

[0180] The display apparatus **100** and image **101** may be set automatically by controller **919** to achieve a certain desired or mandated privacy level specified by policy **240**.

[0181] In one optional step LED driver **715** may be used to adjust display luminance.

[0182] In further optional step colour controller **922** may be used to adjust the colour of the image **101** as illustrated in FIG. **9** for example.

[0183] In further optional step contrast controller **918** may provide reduced image **101** contrast for example by adjustment of image gamma curve to achieve an improved Visual Security Level.

[0184] In further optional step spatial frequency controller **924** may provide image **101** with reduced font sizes, and image blur for large areas of image to reduce visibility of features at snooper viewing distances and achieve an improved Visual Security Level. The spatial frequencies in the image, (in another example by zooming out or reducing the screen magnification), can make the off-axis image more difficult to read as the snooper is typically further from the display apparatus **100** than the primary user.

[0185] The control system **500** may display the currently achieved privacy score or level as a number or as a traffic light colour **282** on indicator **280**.

[0186] It may be desirable to provide some choice to the user for optimum image performance while providing Visual Security Level within policy **240** limits.

[0187] FIG. **11B** illustrates a flowchart for control of a switchable privacy display apparatus **100** by means of calculation of Visual Security Level and user feedback on preferred improvements.

[0188] If the display achieves the correct Visual Security Level after notifying the user of the Visual Security Level in indicator **280** switch **923** provides the current settings to the display, otherwise improvements **921** may be proposed that are based on policy **240** for different types of image **101**. The controller **919** is arranged to provide display luminance, contrast, white points and spatial frequencies to the users' selection. The Visual Security Level is recalculated and the result given to the user. When the performance is adequate in user preference and Visual Security Level the display can proceed to display desired confidential data.

[0189] Thus adjustments to the display apparatus **100** and image **101** may be set automatically or recommendations to the user to set values in order to maintain a certain desired or mandated privacy level based on image preferences for the user. For example, a user may have little concern about the colour of an image but may desire large text size, or a user may prefer a well rendered colour image but with reduced luminance. The choice may depend on the application being used and be different for say images and text. Visual Security Level may be set to provide similar performance for both situations, based on said user preference.

[0190] In a privacy display apparatus **100** the setting of a privacy mode will have some effect on the viewing experience of the primary user, such as narrowing their viewing freedom or comfort. If this effect is more severe than necessary for the primary user, they may be inhibited from using the privacy display apparatus **100**. Advantageously the Visual Security Level may be set at an adequate amount while maintaining minimum changes to the viewing experience of the primary user.

[0191] It may be desirable for the display apparatus **100** to provide adequate Visual Security Level when a sensitive document or information is to be displayed. It may further be desirable for the display apparatus **100** to switch to a privacy mode when sensitive data is entered by the primary user.

[0192] FIG. 11C illustrates a flowchart for control of a privacy display comprising an allowable visual security level for a working document.

[0193] The control system **500** may selectively operate the display apparatus **100** in the public mode or the privacy mode in response to the context of the application software being executed.

[0194] The controller may provide inspection **930** of document security flags, or notification **932** of confidential data entry, for example by noting the use of an appropriate dialog box. The document may also have inspection **934** of document keywords that flag use of privacy operation. The controller may provide inspection **931** of the data transfer security protocol used for example when a web page is accessed. With a secure hypertext transfer protocol for example https is used the display apparatus may be controlled to enter privacy mode. When an open hypertext transfer protocol for example http is used the display apparatus may be controlled to be in public mode.

[0195] The control system **500** may process data sources including if personal data dialog entry **1070** is underway such as a software application requesting a user name or password or a date of birth for example. Based on this information the display apparatus **100** may be switched to a privacy mode automatically during data entry and then switched back to the public mode when such data entry is finished. In this way the user is not troubled to manually switch the display in and out of privacy mode. Thus, for certain documents, the Visual Security Level may be adjusted to maximise privacy achieved in comparison to other less sensitive documents. Such adjustment may be by means of luminance control, contrast, colour and spatial frequency adjustment as described elsewhere herein.

[0196] The acceptable Visual Security Level policy **240** may be adjusted dependent on the application in use. After calculation of Visual Security Level switch **923** may be arranged to allow **938** the document or application to be used or deny **940** the document or application use.

[0197] The display apparatus **100** is capable of executing application software and from the context of that software to recognise if the user is being asked to enter sensitive information such as passwords or personal identity information via for example a dialog box. The display apparatus **100** is capable of executing application software for displaying a document and the display apparatus **100** is arranged to receive user input. In that context the control system **500** may automatically switch the display apparatus **100** in to privacy mode and then automatically switch out of privacy mode when such dialog entry has finished. Typically in a dialog box for a password the typed characters do not appear on the screen so it may not be necessary to switch to privacy mode for this, however the fact that a document may require a password is an indication that the document should be kept secure and that privacy mode should be automatically entered after completion of password entry as the document is opened.

[0198] Further the control system **500** selectively operates the display device in the public mode or the privacy mode in response to the content of the document and/or metadata associated with the document. The control system **500** may inspect the document classification or document security flags **1068** or other meta-data and/or may inspect **934** the document for keywords and process these data sources in order to reliably determine if the document about to be displayed on the screen contains sensitive data and before displaying the image, automatically switch the display apparatus **100** to an appropriate privacy level as required by corporate policy and as appropriate for the location determined by the Network ID **201**, GPS sensor **206** and ambient light level determined by Ambient Light Sensor **232**.

[0199] Alternatively, the application running, for example a screen share initiated from a remote person or a video call from a remote person may request the primary user's display apparatus **100** to enter a privacy mode.

[0200] The display apparatus **100** may be set to a privacy mode before sensitive data is displayed. The display apparatus **100** may switch to a privacy mode when sensitive data is entered by the primary user or when a sensitive document or information is to be displayed. Advantageously the overall vulnerability to data loss from snooping is reduced.



[0201] The control of a privacy display for use with documents comprising entry of a password will now be further described.

[0202] FIG. **11D** illustrates a flowchart for control of a privacy display comprising an allowable visual security level for a password protected document.

[0203] In a first step **950** a user may attempt to open a document. In a second step **952** it is determined whether a password protected document is used. If not, then the privacy settings of the display are determined by other policy **240** requirements of the display system.

[0204] In step **958** if the document is opened then the display is operated in privacy mode with control **919**. After closing the document in step **960**, the privacy operating mode is reset in step **962**.

[0205] Advantageously, the operation in privacy mode may be determined by password protection documents.

[0206] Features of the embodiment of FIG. **11A** and FIG. **11B** and FIG. **11C** and FIG. **11D** not discussed in further detail may be assumed to correspond to the features with equivalent reference numerals as discussed above, including any potential variations in the features.

[0207] The desirable visibility of images in public mode for an off-axis viewer will now be further described.

[0208] FIGS. **12A-12B** are schematic diagrams illustrating a front view of a privacy display comprising a privacy control system for adjustment of image visibility in public mode.

[0209] In a high ambient luminance environment as illustrated in FIG. **12A** it may be desirable to have a high display luminance setting **278** to overcome reflections from ambient illumination **604**. However, in low risk environments, it may still be desirable to provide some low level of privacy and to reduce stray light while achieving increased uniformity to the user in comparison to privacy mode operation with a high Visual Security Level. Such illuminance may reduce the time between charging intervals of the battery.

[0210] The control system **500** of the present embodiments may also achieve controllable performance in public mode of operation as illustrated in FIG. **12B**, for example with reduced ambient illuminance. In such arrangements, it may be desirable to reduce the off-axis luminance to increase display apparatus **100** efficiency.

[0211] Features of the embodiment of FIG. **12A** and FIG. **12B** not discussed in further detail may be assumed to correspond to the features with equivalent reference numerals as discussed above, including any potential variations in the features.

[0212] Further, the output to an off-axis viewer can be adjusted to provide a high quality image with high efficiency as will now be described further.

[0213] FIG. **13** is a schematic graph illustrating variation of image visibility,  $W$  with off-axis relative luminance  $P$  of a switchable privacy display **100** operating in public mode for two different ratios of head-on luminance to illuminance ratios respectively.

[0214] Profile **470** illustrates the variation of image visibility  $W$  with off-axis relative luminance,  $P$  for a ratio of 1 nit head-on luminance per 1 lux of ambient Lambertian illuminance (for example a 300 nit head-on luminance display illuminated by 300 lux) and profile **472** illustrates the variation of  $W$  with  $P$  for 2 nit head-on luminance per 1 lux of ambient Lambertian illuminance (for example a 300 nit head-on luminance display illuminated by 150 lux).

[0215] For desirable wide angle image visibility, a value of  $W$  of greater than 0.85, preferably greater than 0.9 and most preferably greater than 0.95 is desirable for an easily observed image. Displays that desirably achieve high image contrast have contrast sensitivities of greater than 90% and preferably greater than 95%.

[0216] Thus for typical wide angle displays operating in ambient illumination, it is desirable to provide an off-axis luminance such that the image visibility is slowly varying and above a desirable image visibility such as illustrated by line **474** at which a 1 nit/lux display condition provides an image visibility of 0.9. Preferably a switchable privacy display achieves an off-axis relative

luminance for an off-axis viewer (for example at a lateral viewing angle of 45 degrees) of greater than 15% and most preferably an off-axis relative luminance, P of greater than 30% as illustrated by line **475** for an image visibility W of greater than 0.95. As illustrated by line **472**, increasing display luminance to 2 nits/lux provides increased image visibility W, as would be expected.

[0217] Features of the embodiment of FIG. **13** not discussed in further detail may be assumed to correspond to the features with equivalent reference numerals as discussed above, including any potential variations in the features.

[0218] FIG. **14** illustrates a flowchart for control of a switchable privacy display operating in public mode by means of calculation of image visibility for off-axis users.

[0219] After setting **480** of public mode, the ambient light sensor **232** determines ambient illuminance, and illuminance controller **715** can be used to determine display luminance. Optionally contrast, spatial frequency and colour properties of image **101** may be determined by image analysis engine **906**.

[0220] Control system **482** is used to determine image visibility, W for an off-axis user and the display luminance and image contrast is adjusted to provide an appropriate image visibility that overcomes ambient illuminance.

[0221] Advantageously a display with reduced off-axis luminance in public mode may be provided, so that the Visual Security Level when operated in privacy mode is enhanced. Further, the image appearance to an off-axis user in public mode is enhanced.

[0222] Features of the embodiment of FIG. **14** not discussed in further detail may be assumed to correspond to the features with equivalent reference numerals as discussed above, including any potential variations in the features.

[0223] Examples of displays that are capable of switching between privacy mode and a public mode will now be described.

[0224] FIG. **15** is a schematic diagram illustrating a front perspective view a switchable directional display apparatus **100** comprising a backlight **20**, switchable liquid crystal retarder **300** and a spatial light modulator **48**.

[0225] Display device **100** comprises a switchable directional backlight **20** arranged to output light, the backlight **20** comprising a directional waveguide **1**; and plural light sources **15** arranged to input input light into the waveguide **1**, the waveguide **1**, a rear reflector and light control films **5** being arranged to direct light from light sources **15**, **17** into solid angular extent **402A**. Light control films **5** may comprise turning films and diffusers for example.

[0226] In the present disclosure a solid angular extent is the solid angle of a light cone within which the luminance is greater than a given relative luminance to the peak luminance. For example the luminance roll-off may be to a 50% relative luminance so that the solid angular extent has an angular width in a given direction (such as the lateral direction) that is the same as the full-width half maximum (FWHM).

[0227] The backlight **20** may be arranged to provide an angular light solid angular extent **402A** that has reduced luminance for off-axis viewing positions in comparison to head-on luminance.

[0228] Display control system **710** is arranged to provide control of light source driver **715**. Luminance of LEDs **15** may be controlled by control system, such that absolute off-axis luminance to a snooper may be controlled.

[0229] The spatial light modulator **48** may comprise a liquid crystal display comprising substrates **212**, **216**, and liquid crystal layer **214** having red, green and blue pixels **220**, **222**, **224**. The spatial light modulator **48** has an input display polariser **210** and an output display polariser **218** on opposite sides thereof. The output display polariser **218** is arranged to provide high extinction ratio for light from the pixels **220**, **222**, **224** of the spatial light modulator **48**. Typical polarisers **210**, **218** may be absorbing polarisers such as dichroic polarisers.

[0230] Optionally a reflective polariser **208** may be provided between the dichroic input display polariser **210** and backlight **210** to provide recirculated light and increase display efficiency.

Advantageously efficiency may be increased.

[0231] The optical stack to provide control off-axis luminance will now be described.

[0232] Reflective polariser **302**, plural retarders **300** and additional polariser **318** are arranged to receive output light from the spatial light modulator **48**.

[0233] The plural retarders **300** are arranged between the reflective polariser **302** and an additional polariser **318**. The polarisers **210**, **218**, **318** may be absorbing type polarisers such as iodine polarisers while the reflective polariser **302** may be a stretched birefringent film stack such as APF from 3M Corporation or a wire grid polariser.

[0234] Plural retarders **300** comprise a switchable liquid crystal retarder **301** comprising a layer **314** of liquid crystal material, and substrates **312**, **316** arranged between the reflective polariser **302** and the additional polariser **318**. Retarder **300** further comprises a passive retarder **330** as will be described further below.

[0235] As described below, plural retarders **300** do not affect the luminance of light passing through the reflective polariser **302**, the retarders **300** and the additional polariser **318** along an axis along a normal to the plane of the retarders **300** but the retarders **300** do reduce the luminance of light passing therethrough along an axis inclined to a normal to the plane of the retarders **300**, at least in one of the switchable states of the switchable retarder **301**. This arises from the presence or absence of a phase shift introduced by the retarders **300** to light along axes that are angled differently with respect to the liquid crystal material of the retarders **300**.

[0236] Transparent substrates **312**, **316** of the switchable liquid crystal retarder **301** comprise electrodes arranged to provide a voltage across a layer **314** of liquid crystal material **414** therebetween. Control system **752** is arranged to control the voltage applied by voltage driver **350** across the electrodes of the switchable liquid crystal retarder **301**.

[0237] Features of the embodiment of FIG. **15** not discussed in further detail may be assumed to correspond to the features with equivalent reference numerals as discussed above, including any potential variations in the features.

[0238] As will be described further below, the additional polariser **318**, plural retarders **300** and reflective polariser **302** may be arranged to provide polar control of output luminance and frontal reflectivity from ambient illumination **604**.

[0239] The optical stack to provide control of off-axis luminance will now be described.

[0240] FIG. **16A** is a schematic diagram illustrating in perspective side view an arrangement of the plural retarders **300** in a privacy mode of operation comprising a negative C-plate passive retarder **330** and homeotropically aligned switchable liquid crystal retarder **301** in a privacy mode of operation. In FIG. **16A**, some layers of the optical stack are omitted for clarity. For example the switchable liquid crystal retarder **301** is shown omitting the substrates **312**, **316**.

[0241] The electric vector transmission direction of the reflective polariser **302** is parallel to the electric vector transmission direction of the output polariser **218**. Further the electric vector transmission direction **303** of the reflective polariser **302** is parallel to the electric vector transmission direction **319** of the additional polariser **318**.

[0242] The switchable liquid crystal retarder **301** comprises a layer **314** of liquid crystal material **414** with a negative dielectric anisotropy. The passive retarder **330** comprises a negative C-plate having an optical axis perpendicular to the plane of the retarder **330**, illustrated schematically by the orientation of the discotic material **430**.

[0243] The liquid crystal retarder **301** further comprises transmissive electrodes **413**, **415** arranged to control the liquid crystal material, the layer of liquid crystal material being switchable by means of adjusting the voltage being applied to the electrodes. The electrodes **413**, **415** may be across the layer **314** and are arranged to apply a voltage for controlling the liquid crystal retarder **301**. The transmissive electrodes are on opposite sides of the layer of liquid crystal material **414** and may for example be ITO electrodes.

[0244] Alignment layers may be formed between electrodes **413**, **415** and the liquid crystal material

**414** of the layer **314**. The orientation of the liquid crystal molecules in the x-y plane is determined by the pretilt direction of the alignment layers so that each alignment layer has a pretilt wherein the pretilt of each alignment layer has a pretilt direction with a component **417a**, **417b** in the plane of the layer **314** that is parallel or anti-parallel or orthogonal to the electric vector transmission direction **303** of the reflective polariser **302**.

[0245] Driver **350** provides a voltage V to electrodes **413**, **415** across the layer **314** of switchable liquid crystal material **414** such that liquid crystal molecules are inclined at a tilt angle to the vertical. The plane of the tilt is determined by the pretilt direction of alignment layers formed on the inner surfaces of substrates **312**, **316**.

[0246] In typical use for switching between a public mode and a privacy mode, the layer of liquid crystal material is switchable between two states, the first state being a public mode so that the display may be used by multiple users, the second state being a privacy mode for use by a primary user with minimal visibility by snoopers. The switching may be by means of a voltage being applied across the electrodes. In general such a display may be considered having a first wide angle state and a second reduced off-axis luminance state.

[0247] Features of the embodiment of FIG. **16A** not discussed in further detail may be assumed to correspond to the features with equivalent reference numerals as discussed above, including any potential variations in the features.

[0248] The propagation of polarised light from the output polariser **218** will now be considered for on-axis and off-axis directions for a display operating in privacy mode.

[0249] FIG. **16B** is a schematic diagram illustrating in side view propagation of output light from a spatial light modulator through the optical stack of FIG. **15** in a privacy mode of operation; and FIG. **16C** is a schematic graph illustrating the variation of output luminance with polar direction for the transmitted light rays in FIG. **16B**.

[0250] When the layer **314** of liquid crystal material **414** is driven to operate in the privacy mode, the retarders **300** provide no overall transformation of polarisation component **360** to output light rays **400** passing therethrough along an axis perpendicular to the plane of the switchable retarder, but provides an overall transformation of polarisation component **361** to light rays **402** passing therethrough for some polar angles which are at an acute angle to the perpendicular to the plane of the retarders.

[0251] Polarisation component **360** from the output polariser **218** is transmitted by reflective polariser **302** and incident on retarders **300**. On-axis light has a polarisation component **362** that is unmodified from component **360** while off-axis light has a polarisation component **364** that is transformed by the retarders **300**. At a minimum, the polarisation component **361** is transformed to a linear polarisation component **364** and absorbed by additional polariser **318**. More generally, the polarisation component **361** is transformed to an elliptical polarisation component, that is partially absorbed by additional polariser **318**.

[0252] Thus in a polar representation of transmission by the retarders **300** and additional polariser **318** in a privacy mode, regions of high transmission and regions of low transmission are provided as illustrated in FIG. **16C**.

[0253] The polar distribution of light transmission illustrated in FIG. **16C** modifies the polar distribution of luminance output of the underlying spatial light modulator **48**. In the case that the spatial light modulator **48** comprises a directional backlight **20** then off-axis luminance may be further be reduced as described above.

[0254] Features of the embodiment of FIG. **16B** and FIG. **16C** not discussed in further detail may be assumed to correspond to the features with equivalent reference numerals as discussed above, including any potential variations in the features.

[0255] Advantageously, a privacy display is provided that has low luminance to an off-axis snoopers while maintaining high luminance for an on-axis observer. FIG. **16C** further illustrates target snoopers polar location points **27L**, **27R** that may be used for evaluation of Visual Security Level as

will be described further below.

[0256] The operation of the reflective polariser **302** for light from ambient light source **604** will now be described for the display operating in privacy mode.

[0257] FIG. **16D** is a schematic diagram illustrating in top view propagation of ambient illumination light through the optical stack of FIG. **15** in a privacy mode of operation; and FIG. **16E** is a schematic graph illustrating the variation of reflectivity with polar direction for the reflected light rays in FIG. **16D**.

[0258] Ambient light source **604** illuminates the display apparatus **100** with unpolarised light. Additional polariser **318** transmits light ray **410** normal to the display surface with a first polarisation component **372** that is a linear polarisation component parallel to the electric vector transmission direction **319** of the additional polariser **318**.

[0259] In both states of operation, the polarisation component **372** remains unmodified by the retarders **300** and so transmitted polarisation component **382** is parallel to the transmission axis of the reflective polariser **302** and the output polariser **218**, so ambient light is directed through the spatial light modulator **48** and lost.

[0260] By comparison, for ray **412**, off-axis light is directed through the retarders **300** such that polarisation component **374** incident on the reflective polariser **302** may be reflected. Such polarisation component is re-converted into component **376** after passing through retarders **300** and is transmitted through the additional polariser **318**.

[0261] Thus when the layer **314** of liquid crystal material is in the second state of said two states, the reflective polariser **302** provides no reflected light for ambient light rays **410** passing through the additional polariser **318** and then the retarders **300** along an axis perpendicular to the plane of the retarders **300**, but provides reflected light rays **412** for ambient light passing through the additional polariser **318** and then the retarders **300** at some polar angles which are at an acute angle to the perpendicular to the plane of the retarders **300**; wherein the reflected light **412** passes back through the retarders **300** and is then transmitted by the additional polariser **318**.

[0262] The retarders **300** thus provide no overall transformation of polarisation component **380** to ambient light rays **410** passing through the additional polariser **318** and then the retarder **300** along an axis perpendicular to the plane of the switchable retarder, but provides an overall transformation of polarisation component **372** to ambient light rays **412** passing through the absorptive polariser **318** and then the retarders **300** at some polar angles which are at an acute angle to the perpendicular to the plane of the retarders **300**.

[0263] The polar distribution of light reflection illustrated in FIG. **16E** thus illustrates that high reflectivity can be provided at typical snoop locations by means of the privacy state of the retarders **300**. Thus, in the privacy mode of operation, the reflectivity for off-axis viewing positions is increased, and the luminance for off-axis light from the spatial light modulator is reduced as illustrated in FIG. **16C**.

[0264] In the public mode of operation, the control system **710**, **752**, **350** is arranged to switch the switchable liquid crystal retarder **301** into a second retarder state in which a phase shift is introduced to polarisation components of light passing therethrough along an axis inclined to a normal to the plane of the switchable liquid crystal retarder **301**.

[0265] By way of comparison, solid angular extent **402D** may be substantially the same as solid angular extent **402B** in a public mode of operation. Such control of output solid angular extents **402C**, **402D** may be achieved by synchronous control of the sets **15**, **17** of light sources and the at least one switchable liquid crystal retarder **300**.

[0266] Advantageously a privacy mode may be achieved with low image visibility for off-axis viewing and a large solid angular extent may be provided with high efficiency for a public mode of operation, for sharing display imagery between multiple users and increasing image spatial uniformity.

[0267] Additional polariser **318** is arranged on the same output side of the spatial light modulator

**48** as the display output polariser **218** which may be an absorbing dichroic polariser. The display polariser **218** and the additional polariser **318** have electric vector transmission directions **219**, **319** that are parallel. As will be described below, such parallel alignment provides high transmission for central viewing locations.

[0268] A transmissive spatial light modulator **48** arranged to receive the output light from the backlight; an input polariser **210** arranged on the input side of the spatial light modulator between the backlight **20** and the spatial light modulator **48**; an output polariser **218** arranged on the output side of the spatial light modulator **48**; an additional polariser **318** arranged on the output side of the output polariser **218**; and a switchable liquid crystal retarder **300** comprising a layer **314** of liquid crystal material arranged between the at least one additional polariser **318** and the output polariser **318** in this case in which the additional polariser **318** is arranged on the output side of the output polariser **218**; and a control system **710** arranged to synchronously control the light sources **15**, **17** and the at least one switchable liquid crystal retarder **300**.

[0269] Control system **710** further comprises control of voltage controller **752** that is arranged to provide control of voltage driver **350**, in order to achieve control of switchable liquid crystal retarder **301**.

[0270] Features of the embodiment of FIG. **16D** and FIG. **16E** not discussed in further detail may be assumed to correspond to the features with equivalent reference numerals as discussed above, including any potential variations in the features.

[0271] Advantageously, a privacy display is provided that has high reflectivity to an off-axis snoopers while maintaining low reflectivity for an on-axis observer. As will be described below, such increased reflectivity provides enhanced privacy performance for the display in an ambiently illuminated environment.

[0272] FIG. **17A** is a schematic diagram illustrating in perspective side view an arrangement of the retarders **300** in a public mode of operation. In the present embodiment, zero volts is provided across the liquid crystal retarder **301**; and TABLE 2 describes an illustrative embodiment for the arrangement of FIG. **17A**.

TABLE-US-00002 TABLE 2 Passive retarder(s) Active LC retarder  $\Delta n \cdot d$ / Alignment Pretilt/  $\Delta n \cdot d$ / Volt- Mode Type nm layers deg nm  $\Delta \epsilon$  age/V Wide Negative  $-700$  Homeotropic  $88$   $810$   $-4.3$   $0$  Privacy C Homeotropic  $88$   $2.2$

[0273] The switchable liquid crystal retarder **301** comprises two surface alignment layers disposed on electrodes **413**, **415** and adjacent to the layer of liquid crystal material **414** and on opposite sides thereof and each arranged to provide homeotropic alignment in the adjacent liquid crystal material **414**. The layer of liquid crystal material **414** of the switchable liquid crystal retarder **301** comprises a liquid crystal material with a negative dielectric anisotropy. The liquid crystal molecules **414** may be provided with a pretilt, for example  $88$  degrees from the horizontal to remove degeneracy in switching.

[0274] In the present embodiments, desirable ranges for retardations and voltages have been established by means of simulation of retarder stacks and experiment with display optical stacks. Ranges for retardances will now be described that provide design configurations for various optical layers.

[0275] The layer **314** of liquid crystal material has a retardance for light of a wavelength of  $550$  nm in a range from  $500$  nm to  $1000$  nm, preferably in a range from  $600$  nm to  $900$  nm and most preferably in a range from  $700$  nm to  $850$  nm; and the retarder **330** further comprises a passive retarder having an optical axis perpendicular to the plane of the retarder, the passive retarder having a retardance for light of a wavelength of  $550$  nm in a range from  $-300$  nm to  $-900$  nm, preferably in a range from  $-450$  nm to  $-800$  nm and most preferably in a range from  $-500$  nm to  $-725$  nm.

[0276] Features of the embodiment of FIG. **17A** not discussed in further detail may be assumed to correspond to the features with equivalent reference numerals as discussed above, including any potential variations in the features.

[0277] The propagation of polarised light from the output polariser **218** will now be considered for on-axis and off-axis directions for a display operating in public mode for an arrangement of plural retarders **300** as illustrated in FIG. **17A**.

[0278] FIG. **17B** is a schematic diagram illustrating in side view propagation of output light from a spatial light modulator through the optical stack of FIG. **1** in a public mode of operation; and FIG. **17C** is a schematic graph illustrating the variation of output luminance with polar direction for the transmitted light rays in FIG. **17B**.

[0279] Features of the embodiment of FIG. **17B** and FIG. **17C** not discussed in further detail may be assumed to correspond to the features with equivalent reference numerals as discussed above, including any potential variations in the features.

[0280] When the liquid crystal retarder **301** is in a first state of said two states, the retarders **300** provide no overall transformation of polarisation component **360**, **361** to output light passing therethrough perpendicular to the plane of the switchable retarder **301** or at an acute angle to the perpendicular to the plane of the switchable retarder **301**. That is polarisation component **362** is substantially the same as polarisation component **360** and polarisation component **364** is substantially the same as polarisation component **361**. Thus the angular transmission profile of FIG. **17C** is substantially uniformly transmitting across a wide polar region. Advantageously a display may be switched to a wide field of view.

[0281] FIG. **17D** is a schematic diagram illustrating in top view propagation of ambient illumination light through the optical stack of FIG. **1** in a public mode of operation; and FIG. **17E** is a schematic graph illustrating the variation of reflectivity with polar direction for the reflected light rays in FIG. **17D**.

[0282] Thus when the liquid crystal retarder **301** is in the first state of said two states, the retarders **300** provide no overall transformation of polarisation component **372** to ambient light rays **412** passing through the additional polariser **318** and then the retarders **300**, that is perpendicular to the plane of the retarders **300** or at an acute angle to the perpendicular to the plane of the retarders **300**.

[0283] In operation in the public mode, input light ray **412** has polarisation state **372** after transmission through the additional polariser **318**. For both head-on and off-axis directions no polarisation transformation occurs and thus the reflectivity for light rays **402** from the reflective polariser **302** is low. Light ray **412** is transmitted by reflective polariser **302** and lost in the display polarisers **218**, **210** or the backlight of FIG. **1** or optical isolator **218**, **518** in an emissive spatial light modulator **38** of FIG. **2**.

[0284] Features of the embodiment of FIG. **17D** and FIG. **17E** not discussed in further detail may be assumed to correspond to the features with equivalent reference numerals as discussed above, including any potential variations in the features.

[0285] Advantageously in a public mode of operation, high luminance and low reflectivity is provided across a wide field of view. Such a display can be conveniently viewed with high contrast by multiple observers.

[0286] Other types of switchable privacy display will now be described.

[0287] A display apparatus **100** that may be switched between privacy and public modes of operation comprises an imaging waveguide and an array of light sources as described in U.S. Pat. No. 9,519,153, which is incorporated by reference herein in its entirety. The imaging waveguide images an array of light sources to optical windows that may be controlled to provide high luminance on axis and low luminance off axis in a privacy mode, and high luminance with a large solid angle cone for public operation.

[0288] Switchable angular contrast profile liquid crystal displays are described in Japanese Patent Publ. No. JPH1130783 and in U.S. Patent Publ. No. 2017-0123241, both of which are incorporated by reference herein in their entireties. Such displays may provide out-of-plane tilt of liquid crystal molecules in the liquid crystal layer **214** of a liquid crystal display and may achieve reduced off-axis image contrast in privacy mode of operation. The display apparatus **100** control system **500**

may further comprise control of out-of-plane tilt of the liquid crystal molecules.

[0289] As may be used herein, the terms “substantially” and “approximately” provide an industry-accepted tolerance for its corresponding term and/or relativity between items. Such an industry-accepted tolerance ranges from zero percent to ten percent and corresponds to, but is not limited to, component values, angles, et cetera. Such relativity between items ranges between approximately zero percent to ten percent.

[0290] While various embodiments in accordance with the principles disclosed herein have been described above, it should be understood that they have been presented by way of example only, and not limitation. Thus, the breadth and scope of this disclosure should not be limited by any of the above-described exemplary embodiments, but should be defined only in accordance with any claims and their equivalents issuing from this disclosure. Furthermore, the above advantages and features are provided in described embodiments, but shall not limit the application of such issued claims to processes and structures accomplishing any or all of the above advantages.

[0291] Additionally, the section headings herein are provided for consistency with the suggestions under 37 CFR 1.77 or otherwise to provide organizational cues. These headings shall not limit or characterize the embodiment(s) set out in any claims that may issue from this disclosure.

Specifically and by way of example, although the headings refer to a “Technical Field,” the claims should not be limited by the language chosen under this heading to describe the so-called field.

Further, a description of a technology in the “Background” is not to be construed as an admission that certain technology is prior art to any embodiment(s) in this disclosure. Neither is the “Summary” to be considered as a characterization of the embodiment(s) set forth in issued claims.

Furthermore, any reference in this disclosure to “invention” in the singular should not be used to argue that there is only a single point of novelty in this disclosure. Multiple embodiments may be set forth according to the limitations of the multiple claims issuing from this disclosure, and such claims accordingly define the embodiment(s), and their equivalents, that are protected thereby. In all instances, the scope of such claims shall be considered on their own merits in light of this disclosure, but should not be constrained by the headings set forth herein.

## Claims

**1-22.** (canceled)

**23.** A display apparatus comprising: a display device arranged to display an image and capable of operating in at least a first mode and a second mode, wherein the image remains visible in both the first mode and the second mode in an on-axis direction, and a visibility of the image is reduced in an off-axis direction in the second mode compared to the first mode; and a control system arranged to control the display device, the control system being capable of selectively operating the display device in the first mode or the second mode, and wherein the control system is arranged to selectively control luminance of the displayed image in the first mode based on a first function that relates a level of luminance to a detected level of ambient light and to selectively control luminance of the displayed image in the second mode based on a second function that relates a level of luminance to a detected level of ambient light in the second mode.

**24.** The display apparatus of claim 23, wherein the display apparatus is arranged to connect to a network and the control system is arranged to selectively operate the display device in the first mode or the second mode based on a characteristic associated with the network.

**25.** The display apparatus of claim 23, wherein the control system is arranged to selectively operate the display device in the first mode or the second mode based on an identity of a user.

**26.** The display apparatus of claim 23, wherein the control system is arranged to selectively operate the display device in the first mode or the second mode based on a detected location.

**27.** The display apparatus of claim 23, wherein the display apparatus is arranged to receive a user input and the control system is arranged to selectively operate the display device in the first mode



or the second mode based on the received user input.

**28.** The display apparatus of claim 23, wherein the control system is arranged to selectively operate the display device in the first mode or the second mode based on the detected level of the ambient light.

**29.** The display apparatus of claim 23, wherein the control system is arranged to selectively control at least one of the luminance, a contrast, a white point, or spatial frequency of the displayed image to maintain a selected privacy level based on the detected level of the ambient light.

**30.** The display device of claim 23, wherein the first function relates a higher level of luminance to a detected level of the ambient light than the second function.

**31.** The display apparatus of claim 23, wherein the second function maintains a relationship  $Y_0/I_0 \geq 1$  lux/nit where  $Y_0$  is the maximum output luminance of the display apparatus and  $I_0$  is the detected ambient light level.

**32.** The display apparatus of claim 23, wherein the second function maintains a relationship  $Y_0 \leq Y_{0,\text{sub.lim}}$ , where  $Y_0$  is the maximum output luminance of the display apparatus and  $Y_{0,\text{sub.lim}}$  is given by the equation:  $Y_{0,\text{lim}} = \frac{R_\theta + V_{\text{lim}} * K_\theta}{(V_{\text{lim}} - 1) * P_\theta}$  where  $V_{\text{sub.lim}}$  has a value of 10,  $R_\theta$  is the reflected ambient illuminance at an observation angle  $\theta$  of 45 degrees laterally and zero degrees in elevation from the normal to the display apparatus,  $K_\theta$  is the display black state luminance at the observation angle, and  $P_\theta$  is the relative luminance at the observation angle  $\theta$  compared to the maximum display output luminance  $Y_0$ .

**33.** The display apparatus of claim 23, wherein the display apparatus is arranged to receive a time data input and an ambient light sensor input, and the control system is arranged to selectively operate the display device in the first mode or the second mode based on the time data input and the ambient light sensor input to reduce light pollution generated by the display apparatus.

**34.** A display apparatus of claim 23, wherein the display apparatus is arranged to execute application software and the control system is arranged to selectively operate the display device in the first mode or the second mode based on the application software being executed.

**35.** The display apparatus of claim 34, wherein the display apparatus is arranged to execute application software for displaying a document, and the control system is arranged to selectively operate the display device in the first mode or the second mode based on at least one of a content of the document or a metadata associated with the document.

**36.** The display apparatus of claim 35, wherein at least one of the content or the metadata includes a security status of a transfer protocol of the document, and the transfer protocol is one of Hypertext Transfer Protocol or Hypertext Transfer Protocol Secure.

**37.** The display apparatus of claim 23, where the first mode is a public mode of operation and the second mode is a privacy mode of operation.

**38.** A display apparatus comprising: a display device arranged to display an image and capable of operating in at least a first mode and a second mode, wherein the image remains visible in both the first mode and the second mode in an on-axis direction, and a visibility of the image is reduced in an off-axis direction in the second mode compared to the first mode; and a control system arranged to control the display device, the control system being capable of selectively operating the display device in the first mode or the second mode, and wherein the control system is arranged to selectively control luminance of the displayed image in the first mode based on a first function and to selectively control luminance of the displayed image in the second mode based on a second function.

**39.** The display apparatus of claim 38, wherein the first mode is a public mode of operation and the second mode is a privacy mode of operation.

**40.** The display apparatus of claim 38, wherein the control system is arranged to selectively operate the display device in the first mode or the second mode based on at least one of a characteristic associated with a network that the display apparatus is connected to, an identity of a user, a location of the display apparatus, or a received user input.

- 41.** The display apparatus of claim 38, wherein the control system is arranged to selectively control at least one of the luminance, a contrast, a white point, or spatial frequency of the displayed image to maintain a selected privacy level based on the detected level of the ambient light
- 42.** The display apparatus of claim 38, wherein the first function relates a higher level of luminance to a detected level of the ambient light than the second function.
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